# Cost-Aware High-Fidelity Entanglement Distribution and Purification in the Quantum Internet

Huayue Gu, *Graduate Student Member, IEEE*, Zhouyu Li, *Graduate Student Member, IEEE*, Xiaojian Wang, *Graduate Student Member, IEEE*, Dejun Yang, *Senior Member, IEEE*, Guoliang Xue, *Fellow, IEEE*, and Ruozhou Yu, *Senior Member, IEEE*

***Abstract*—Operating a quantum network incurs high capital and operational expenditures, which are expected to be compensated by the high value of enabled quantum applications. However, existing mechanisms mainly focus on maximizing the entanglement distribution rate and neglect the cost incurred on users. This paper aims to address how to utilize quantum network resources in a cost-efficient manner while sustaining high-quantity and high-quality entanglement distribution. We first consider how to establish a steady stream of entanglements between remote nodes with the minimum cost. Utilizing a recent flow-based abstraction and a novel graph representation, we design an optimal algorithm for min-cost remote entanglement distribution. Next, we consider distributing entanglements with the highest fidelity subject to a cost bound and prove its NP-hardness. To explore the cost-fidelity trade-off due to swapping and purification, we propose an approximation scheme for maximizing fidelity while satisfying an arbitrary cost bound. Our algorithms provide rigorous tools for supporting high-performance quantum network applications with financial consideration and offer strong theoretical guarantees. Extensive simulation results validate the advantageous performance in cost efficiency and/or fidelity compared to existing solutions and heuristics.**

***Index Terms*—Quantum network, entanglement distribution, entanglement purification, fidelity, approximation scheme.**

## I. Introduction

LONG-DISTANCE entanglement distribution is a cornerstone for many promising applications, including quantum key distribution (QKD) [6], remote quantum sensing [21], and distributed quantum computing (DQC) [14]. Recently, notable progress has been made in lab-scale quantum networks [28], [56], [60], hinting at potential large-scale deployment in the near future [31], [33], [71].

The primary challenges of large-scale quantum networking are distance and noise. Entangled photons undergo high loss when transmitted in media over long distances, which cannot be compensated by classical amplification due to the no-cloning theorem [11]. Meanwhile, noisy quantum channels and imperfect quantum operations lead to decoherence of the entangling correlation, which cannot be corrected by classical error correction [52]. To counter these effects, the first-generation quantum repeaters emphasize the use of near-term feasible quantum operations, including entanglement swapping and purification, to distribute entanglement between remote nodes over long distances. Extensive study has been conducted on maximizing the quantity (rate) and quality (fidelity) of entanglement distribution, based on the unique characteristics of these operations [18], [63], [72], [74], [75].

Compared to entanglement rate and fidelity, other factors are less considered yet also critical. While lab-scale testbeds are feasibility-driven and largely cost-agnostic, a commercial quantum network service (such as EBP quantum network [27]) must be built atop a sustainable business model considering the capital and operational expenditures of the service and revenue. For example, deploying a single repeaterless quantum node costs over $200,000 today [1], [3], [27], and the cost for future quantum repeaters with transduction device, quantum memory, and purification is destined to multiply. This is not to mention the costs for workforce development, maintenance and operation. These costs can be compensated by the high value of quantum-enabled applications. Correspondingly, like the internet nowadays, a real quantum internet can be built as a shared infrastructure serving as diverse quantum applications and as many quantum users as possible. Quantum network operators peer with each other to provide global quantum connectivity and decide their service prices based on market trends driven by supply and demand. Users choose among operators based on prices and service quality, and are charged based on services they receive or resources they consume.

While the actual cost of operating a large-scale commercial quantum network is still hard to decide due to technology pre-maturity, we believe it is proper to start investigating

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Huayue Gu, Zhouyu Li, and Ruozhou Yu are with NC State University, Raleigh, NC 27606 USA (e-mail: hgu5@ncsu.edu; zli85@ncsu.edu; ryu5@ncsu.edu).

Xiaojian Wang is with the University of Colorado Denver, Denver, CO 80204 USA (e-mail: xiaojian.wang@ucdenver.edu).

Dejun Yang is with Colorado School of Mines, Golden, CO 80401 USA (e-mail: djyang@mines.edu).

Guoliang Xue is with Arizona State University, Tempe, AZ 85281 USA (e-mail: xue@asu.edu).

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cost management in quantum networking from an *algorithmic perspective*. Consider a user requesting quantum connectivity service in a quantum network. The joint interest of the user and the operator is that the service is fulfilled with the most cost-efficient resources in the network. Nevertheless, because of unique quantum characteristics, the use of different resources (e.g. fiber bandwidth, quantum storage, swapping, and purification devices) has a profound impact on both the cost and the quality of the entanglement distribution service, including its rate and achievable fidelity. In this case, exploring the trade-off between cost and rate/fidelity of entanglement distribution, and finding the most cost-efficient way of serving a user request, become a highly non-trivial problem. As a first-step towards combining quantum network and internet economics, this paper focuses on studying two problems from an algorithmic perspective: (i) finding the *minimum-cost* way of establishing a stream of remote entanglements with a steady expected rate, and (ii) establishing a stream of *high-fidelity* entanglements subject to the user's cost bound. Our main contributions are:

- We formulate min-cost and cost-aware high-fidelity entanglement distribution problems in a quantum network, which, to our best knowledge, have not been studied nor solved with theoretical guarantees in the literature.
- We develop a Purification and Swapping Graph (PSG) abstraction, based on which we design the first optimal algorithm for the min-cost distribution of a steady stream of entanglements between remote nodes.
- We prove NP-hardness of the cost-aware high-fidelity entanglement distribution problem and then propose the first fully polynomial-time approximation scheme (FPTAS) for the problem based on an extended PSG abstraction.
- We conduct extensive simulations and demonstrate that the proposed algorithms achieve high cost efficiency and/or fidelity when achieving the same rate as existing works.

**Organization:** §II presents the background on classical and quantum internet economics. §III shows quantum preliminaries and network model. §IV presents an optimal algorithm for min-cost entanglement distribution. §V presents an approximation scheme to maximize fidelity with a cost bound. §VI presents evaluation results. §VII surveys the related work. §VIII and §IX presents the discussion and conclusion.

## II. Classical and Quantum Internet Economics

In this section, we present classical and quantum Internet economics, respectively. We also emphasize the differences between these two economics.

### A. Classical Internet Economics

The economic model of the classical Internet serves as an excellent example of how the quantum internet ecosystem can evolve. Here, we briefly review the cost and profit model of a classical Internet Service Provider (ISP).

**ISP cost model:** An ISP incurs both *capital expenditure (CapEx)* and *operational expenditure (OpEx)*. The costs for deploying network devices and materials (wire/fiber), land usage, construction, labor and permitting, and research and engineering development constitute major CapEx of an ISP. Further, operating a network infrastructure incurs continuous energy, maintenance, upgrade, device/material/space renting, customer support, as well as costs associated with peering with other ISPs to provide global connectivity to its customers, which constitute the OpEx. Both costs can be substantial on a yearly basis due to the ISP's need to continuously upgrade and operate the infrastructure for market competitiveness. For instance, the US broadband providers incurred over \$102.4 billion in CapEx, marking a 19% year-to-year increase [2].

**ISP profit model:** ISPs subsidize the costs and make profit by charging for Internet users' usage of their infrastructure resources. A residential user is typically charged monthly based on a pre-defined service category (best-effort service with data or data rate cap). For business users, pricing is negotiated case-by-case. Specifically, many business users (such as banking, gaming, or critical services) demand guaranteed service quality such as data rate and/or delay, which must be satisfied by reserved or prioritized resources such as link bandwidth dedicated to each user. In this case, pricing is typically based on the amount of resources that a user's service consumes. In a market with multiple ISPs, a user would choose the most economical service that satisfies its demand. Thus to maximize revenue, an ISP competes by delivering the least price to satisfy a user's demand that can sustain its business, for instance, by leveraging the most cost-efficient resources and optimizing resource utilization. This has motivated extensive research on cost-efficient network optimization in the Internet, such as traffic engineering [73], resource allocation [35], and so on.

### B. Quantum Internet Economics

We expect the future quantum internet ecosystem to be driven by supply and demand similar to those in the classical Internet. Quantum network operators build and maintain expensive quantum network infrastructures such as optical switching centers, repeater facilities or even satellite constellations. These infrastructures will support diverse quantum applications and users to maximize the operators' profit. Users seek competitive prices on demanded services, and operators seek to optimize network operations to reduce cost and improve revenue. Nevertheless, the fundamental differences between classical and quantum networks introduce unique features of the cost management problem in the quantum internet.

**Quantum network service model:** A quantum network provides the basic entanglement distribution service between two or more remote end-points based on demands from different quantum users. Entanglement is a basic resource for all quantum applications. The quantity (i.e. entanglement distribution rate and success probability) and quality (i.e. fidelity) of entanglement distribution fundamentally decide the practicality, usability and performance of supported applications.

In the near future, the quantum network will likely only serve business-level customers and support high-value applications such as high-stake secure communication, distributed quantum computing centers, or highly sensitive sensing in scientific and military domains. These applications typically have high performance requirements regarding entanglement distribution rate and/or fidelity. For instance, distributed quantum computing typically requires abundant high-fidelity entanglement resources to be constantly available between two or more remote quantum computers so that remote quantum operations (such as CNOT gates) can be executed as soon as possible before qubit decoherence [29], [48]. With near-term quantum technology, this is best achieved by maintaining a steady stream of entanglements between any two or more computers, such that purification can be continuously applied to keep entanglement resources at high fidelity in short-lived quantum memories [12], [15], [23]. Similarly, sensors in a quantum sensor network need to maintain steady entanglement connectivity for continuous monitoring on a target quantity [36], [76], and entanglement-based QKD relies on a stream of entanglements between key distribution centers to continuously generate enough keys to serve local users [13], [39], [59], [67].

To provide the performance guarantee, the quantum network infrastructure can act as a virtualization platform, reserving resources to satisfy the need of each user [57], [72]. Each user's service is implemented on dedicated or prioritized resources to guarantee rate and/or fidelity. Correspondingly, users are charged based on **overall resource consumption** in the network for their services, such as fiber bandwidth and quantum (optical) processing at nodes. Each resource may be priced differently based on factors such as current utilization level, deployment/maintenance costs, energy consumption, and so on. To serve a user with the minimum cost/price, it is thus important to **select the most cost-efficient resources** that can satisfy the user's demand.

**Unique features of quantum network cost management:** As we model in detail in the next section, the characteristics of quantum networks make cost management drastically different from the classical Internet.

First, compared to the classical Internet, where bandwidth is a uniform resource across links, quantum resources such as fiber bandwidth exhibit drastic heterogeneity in resource (cost) efficiency for realizing the same service quality. Specifically, factors such as distance may significantly affect the achievable rate and/or fidelity of entanglements using the same optical bandwidth. The success probability of key operations like swapping and purification may also differ across nodes, as decided by the physical environment and device heterogeneity.

Second, each unit resource may result in different rate and/or fidelity when distributing between different end points, as decided by the different swapping and purification processes to implement the end-to-end distribution. A link that can generate 20 entanglements per second (eps) may support only 10 end-to-end eps if one swapping step with success probability 50% is applied, or only 5 end-to-end eps if two swapping steps are applied. This means the length of a path can result in not just linear accumulation of node and link costs, but also multiplicative increase in each node/link's resource consumption to provide a certain entanglement rate or fidelity.

Combined with the non-trivial modeling of processes such as swapping and purification themselves, the overall resource cost minimization problem becomes highly complex, and fundamentally different from cost minimization in classical networks. New theoretical models and algorithmic frameworks are needed to enable efficient cost management in the quantum internet.

TABLE I
NOTATION TABLE OF NETWORK MODEL

| Parameters | Description |
|---|---|
| $G = (V, E)$ | quantum network with nodes $V$ and links $E$ |
| $mn, M$ | an enode, and set of enodes $M = \{mn \mid m, n \in V\}$ |
| $q_e, q_n$ | ebit generation & swapping success probabilities |
| $c_e^{\mathsf{gen}}, c_e^{\mathsf{pur}}, c_n$ | ebit generation, purification & swapping costs |
| $c_e(K)$ | expected generation & $K$-round purification cost on $e$ |
| $c_{st}$ | the long-term expected cost for a given pflow |
| $F_e$ | initial fidelity of elementary ebits generated on link $e$ |
| $F^{\mathsf{E2E}}$ | end-to-end fidelity of a successfully distributed ebit |
| $y_F(K), \phi_F(K)$ | expected yield and fidelity for $K$-round purification from initial fidelity $F$, defined in Eqs. (4)–(5) |
| $W_e, W_n$ | fidelity parameters on link $e$ & swapping at $n$ |
| $W_e^{[K]}$ | fidelity parameter of $K$-round purified ebits on $e$ |
| $\mathbf{K}$ | maximum rounds of link-level purification |
| $p$ | a pflow $\{f_{mn}^{mk}, g_{mn}, \kappa_{mn}\}$ following Definition 1 |
| $F_p, c_p$ | fidelity and expected cost of a pflow $p$ in Eqs. (7)–(8) |
| $I_{mn}$ | total ebit rate generated between node pair $mn$ |
| $\Omega_{mn}$ | total ebit rate contributed by $mn$ to swapping |
| $\lambda_{st}, c_{st}$ | expected EDR and cost of an SD pair in Eqs. (6)–(9) |
| $\Delta_{st}$ | cost bound of an SD pair $st$ |
| $\Upsilon_{st}$ | end-to-end fidelity bound between $s$ and $t$ |
| $\theta$ | quantization factor of path lengths |
| $Z, Z^*$ | length bounds & optimal solution before quantization |

## III. Quantum Preliminaries

In this section, we present preliminaries of a quantum network. Notations related to modeling are summarized in Table I.

### A. Entanglement Distribution Process

We consider the fundamental task of distributing two-qubit (bipartite) entangled states between remote end points. A *maximally entangled bipartite state* (called a Bell state or an *ebit*) is a fundamental resource in quantum communication, which can be used to construct arbitrary multi-partite entangled states and support distributed quantum applications. We focus on distributing one of the four Bell states and use $|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$ as the example[1]. Below, we introduce the physical processes to implement remote entanglement distribution.

**Entanglement generation:** Assume two quantum nodes intend to exchange quantum information. One node with an

[1]All four Bell states—$|\Phi^{\pm}\rangle$ and $|\Psi^{\pm}\rangle$— are symmetric under local operations and classical communication (LOCC).

entanglement source can continuously generate entangled photon pairs via spontaneous parametric down-conversion (SPDC) [16] or four-wave mixing [42] and send one photon of the pair to the other node via a quantum channel such as optical fiber or free space. Alternatively, a dedicated entanglement source can reside in-between the two nodes, and both photons of a generated pair can be transmitted to the two nodes respectively along the fiber. This is entanglement generation, and ebits generated and distributed along a physical channel are *elementary ebits*.

As a non-linear optical operation, the efficiency of the SPDC-based entanglement source is very low with current technologies, typically in the order of $10^{-6}$ ebits per input pump photon [9]. Meanwhile, photon transmission over fiber incurs the exponential loss of photons, typically in the order of 0.1dB/km, leading to excessive loss over tens of kilometers [20]. One way to improve the rate is to increase the pump power [8], [30]. However, this also leads to multi-photon coincidence events—generating more than one entangled pair in one photon detection period that cannot be distinguished—which may even degrade the generation rate [44]. Since the entangled photons will get lost during transmission due to channel attenuation and other environmental factors, the entanglement generation can be viewed as a probabilistic process [55], and we use $q_e \in (0,1]$ to denote the entanglement success probability while considering the transmission loss, the number of attempts in unit time, and the generation efficiency of the entanglement source (see Appendix).

**Entanglement swapping:** Quantum repeaters are designed to alleviate the excessive loss of entangled photons over long distances via entanglement swapping. Assume two remote nodes want to exchange quantum information, instead of directly transmitting entangled photons, both nodes may each establish an ebit with an intermediate repeater, who then concatenates the two ebits into one end-to-end ebit between the two nodes. Each quantum swapping consists of a two-qubit operation for locally entangling the two ebits at the repeater, a one-qubit operation for correction, and a Bell State Measurement (BSM). However, each entanglement swapping operation has a chance to fail due to the protocol design and the imperfections in the swapping circuit [5]. Therefore, we use $q_n \in (0,1]$ to denote the swapping success probability of each repeater $n$.

**Noise and fidelity:** Noise affects the quality of the generated entanglement (turning pure quantum states into *mixed states*). Consider $|\Phi^+\rangle$ as our desired entangled state. To consider the impact of noise, we model an arbitrary noise-affected entanglement (a mixed state) as the Werner state $w_F = F|\Phi^+\rangle\langle\Phi^+|+\frac{1-F}{3}(|\Phi^-\rangle\langle\Phi^-|+|\Psi^+\rangle\langle\Psi^+|+|\Psi^-\rangle\langle\Psi^-|)$ with *fidelity* $F$, which can be transformed from any mixed state with the same fidelity via *random bilateral rotations* (RBR) [7]. This corresponds to a worst-case isotropic error model [58].

Following the derivation in [25], we consider the noise in entanglement generation by defining parameters $W_e \triangleq \frac{4F_e-1}{3}$ and $W_n \triangleq o_{1,n}o_{2,n}\frac{4\alpha_n^2-1}{3}$ for each link and node respectively,

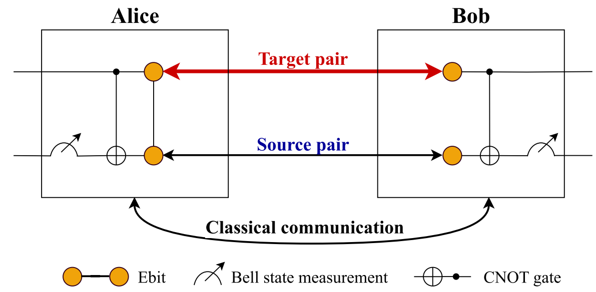



Fig. 1. Entanglement purification to improve the fidelity of a target pair by consuming a source pair between two parties.

where $F_e$ is the initial fidelity of elementary ebits generated along link $e$, and $o_{1,n}, o_{2,n}$ are the probabilities of 1-qubit, 2-qubit operations and $\alpha_n$ is the accuracy of the bell state measurement at node $n$. The end-to-end fidelity of a successfully distributed ebit via a path $p = (s, n_1, n_2, \ldots, n_X, t)$ along links $\{e_1, e_2, \ldots, e_{X+1}\}$ is:

$$F^{\mathsf{E2E}} = \frac{1}{4}\cdot\left(1+3\prod_{i=1}^{X+1}W_{e_i}\prod_{j=1}^{X}W_{n_j}\right). \tag{1}$$

**Entanglement purification:** Low fidelity of entanglements limits their applicability in various quantum applications and affects the performances of applications. To counter the effect of noise, entanglement purification can be employed to increase ebit fidelity at the cost of lower ebit rate. As is shown in Figure 1, Alice and Bod hold two entangled qubit pairs, including a target pair and a source pair. Both Alice and Bob first perform the CNOT gate to entangle their local qubits and then measure the source pair in the Bell basis, and compare results via classical communication. If the results are consistent, the target pair now has a higher fidelity towards the desired state with the cost of the source pair. Otherwise, both ebits will be discarded, and the purification will fail. After one round of successful purification, the target ebit can be further purified by using more source ebits. This is called the **recurrence method** for entanglement purification as in [22] and [53].

Given two Werner states with fidelity $F_i$ and $F_j$, the success probability for entanglement purification is [26]:

$$\begin{aligned} q_{ij} = q(F_i,F_j) &\triangleq \left(F_i+\frac{1-F_i}{3}\right)\left(F_j+\frac{1-F_j}{3}\right) \\ &+\frac{2\,(1-F_i)}{3}\cdot\frac{2\,(1-F_j)}{3} = \frac{8F_iF_j-2(F_i+F_j)+5}{9}. \end{aligned} \tag{2}$$

If the purification succeeds, the resulting fidelity is:

$$\begin{aligned} F_{ij} = \varphi(F_i,F_j) &\triangleq \frac{F_iF_j+\frac{(1-F_i)(1-F_j)}{9}}{q(F_i,F_j)} \\ &= \frac{10F_iF_j-(F_i+F_j)+1}{8F_iF_j-2(F_i+F_j)+5}. \end{aligned} \tag{3}$$

Based on the above, we can derive the *yield* $y_F(K)$ (the expected number of target ebits per source ebit) and the fidelity $\phi_F(K)$ after $K$ rounds of purification from initial fidelity $F$:

$$y_F(K) = \frac{q(F, \phi(K-1)) \cdot y_F(K-1)}{y_F(K-1)+1}, \qquad y_F(0) = 1; \tag{4}$$

$$\phi_F(K) = \varphi(F, \phi_F(K-1)), \qquad \phi(0) = F. \tag{5}$$

We define the link-level fidelity parameter after $K$ purification rounds as $W_e^{[K]} \triangleq \frac{4\phi_{F_e}(K)-1}{3}$. As purification requires holding the target ebit in memory for a time period (waiting for $K$ successfully generated source ebits), we define an upper bound $\mathbf{K}$ on the total rounds of purifications for an ebit on a link.

### B. Cost Economics for Entanglement Distribution

As discussed in Sec. II, our pricing model is motivated by the unprecedented success of Internet economics, where *traffic* is the primary commodity sold by Internet Service Providers (ISPs) and eventually consumed by Internet users [43], [46], [65]. The bandwidth on links, along with possible processing on nodes (such as middleboxes), is priced as *resource costs* to the users. In a quantum network, users will similarly be charged based on the amount of resources consumed on nodes and links, including entanglement source capacity, optical bandwidth, and swapping and purification operations. Below, we define the resource cost for each physical process above.

**Entanglement generation:** Consider $c_e^{\mathsf{gen}} > 0$ as the cost for *attempting* to generate one entanglement per unit time along a link $e$. The cost includes using the entanglement source capacity and occupying the optical bandwidth on this link. Considering the success probability of entanglement generation $q_e$, if the target expected rate of generated entanglements over $e$ is $\lambda_e$, then the expected cost to achieve this rate is $c_e^{\mathsf{gen}} \cdot \lambda_e / q_e$, factoring in the success probability on this link.

**Entanglement swapping:** Entanglement swapping consumes quantum memories and quantum operations on a repeater. Because swapping also succeeds only with a probability, if an expected rate of successful swapping at node $n$ is $\lambda_n$, then the expected cost of entanglement swapping to achieve this rate is $c_n \cdot \lambda_n / q_n$, given unit swapping cost $c_n > 0$.

**Entanglement purification:** Purification also consumes memories and quantum operations but is more complicated since multiple rounds of purification may be performed before swapping with the aim of improving the fidelity to the desired one. Define $c_e^{\mathsf{pur}} > 0$ as the cost for performing one purification round, regardless of whether it is successful or not. To generate one ebit after $K$ rounds of successful purification, the expected number of original ebits needed (with initial fidelity $F$) is $1/y_F(K)$, and the expected total rounds of purification (counting in success probability) is $1/y_F(K) - 1$. The expected cost for successfully generating one ebit after $K$ rounds of generation and purification is $c_e(K) = (c_e^{\mathsf{gen}}/q_e + c_e^{\mathsf{pur}})/y_{F_e}(K) - c_e^{\mathsf{pur}}$, with $c_e(0) = c_e^{\mathsf{gen}}/q_e$ consistent with generation without purification as above.

As discussed in Sec. II-B, purchasing different "bundles" of resources has a much more profound impact on the cost-performance trade-off of a user than on the Internet. If a user's service is implemented along paths that result in low rate or low fidelity, then achieving satisfactory rate/fidelity (explained in the next subsection) may cost much more resources than picking some better paths. How to pick the best suite of resources, and utilize them to achieve the best performance is thus an important algorithmic problem faced by a quantum user and/or the network operator. As the future quantum internet is expected to serve many users (SD pairs) with sufficient capacity, this paper focuses on a single SD pair, with possible extension to the multi-SD pair case discussed in Sec. VIII.

### C. Quantum Network Model

We model a quantum network as an undirected graph $G = (V, E)$, where $V$ is the set of quantum nodes, and $E$ is the set of physical channels (links) between nodes. Each link $e \in E$ is equipped with an entanglement generation probability $q_e \in (0, 1]$ and a unit cost $c_e^{\mathsf{gen}} \in \mathbb{R}^+$ for attempting to generate an elementary ebit. Each repeater $v \in V$ is equipped with a swapping success probability $q_v \in (0, 1]$ and a unit cost $c_n \in \mathbb{R}^+$ for swapping. We also use *mn* to denote an *unordered* node pair $\{m, n\}$ for $m, n \in V$, and hence $mn = nm$. Each pair *mn* is called an *enode* denoting a pair of nodes between which entanglement may be established. The set of all enodes is $M = \{mn \,|\, m, n \in V\}$, and note that $E$ is a subset of $M$. Following the conventional notation, we call a pair of communicating nodes *st* a source-destination (SD) pair without explicitly defining which node is the source or destination.

We adopt a time-slotted system model following [74] and [75], while all our definitions and algorithms can be trivially extended to continuous-time asynchronous operations [70]. We also assume a central controller controls all operations in the quantum network as [63] and [75] to monitor network status, allocate resources, and decide the costs of different operations. In each time slot, the following phases are carried out in order:

1) **Entanglement generation:** For a pair of nodes $mn \in E$ with a direct link, they will attempt to generate elementary *mn*-ebits at a pre-defined rate.
2) **Entanglement purification:** Low-fidelity elementary ebits can be purified before being involved in further processes such as swapping. To purify, nodes $mn \in E$ will utilize a pre-defined number of generated low-fidelity elementary ebits to generate a high-fidelity ebit.
3) **Entanglement swapping:** When ebits are available between enode *mk* and *kn* sharing the repeater $k$, repeater $k$ can attempt to perform swapping between pairs of *mk*- and *kn*-ebits to create ebits between nodes *mn*. The source *mk*- and *kn*-ebits are not required to be elementary.

### D. Satisfying User Demand

As motivated in Sec. II-B, a user submits its service demand as the expected rate (and/or fidelity) of a steady entanglement stream between an SD pair *st* for an extended period of time. To satisfy this demand, we utilize (and later extend) an abstraction originally developed in [18] to describe how a user demand can be implemented in a quantum network.

Let $\lambda_{st}(\tau)$ be the number of generated *st*-ebits in time slot $\tau$. Let $\mathbb{E}[\cdot]$ denote expectation. The long-term entanglement distribution rate (EDR) is defined as

$$\lambda_{st} = \liminf_{T\to\infty} \frac{1}{T}\sum_{\tau=1}^{T} \mathbb{E}[\lambda_{st}(\tau)]. \quad (6)$$

To achieve an EDR, a remote entanglement distribution (RED) algorithm needs to decide how to generate, purify and swap entanglements in the network. The *(primitive) entanglement flow (pflow)* abstraction in [32] lends us a tool to define and devise an RED algorithm that achieves a long-term EDR $\lambda_{st}$. Since the original definition does not consider purification, our first contribution is to extend the pflow abstraction in [32]:

*Definition 1 (Pflow):* Given quantum network $G$ and SD pair $st \in M$, a pflow with EDR $\lambda_{st}$ is defined by variables $\{g_{mn} \geq 0, \kappa_{mn} \in \mathbb{Z}^* \,|\, mn \in M\} \cup \{f_{mn}^{mk} \geq 0 \,|\, m,k,n \in V\}$, such that

(i) **Generation:** $g_{mn} = \kappa_{mn} = 0$ for $\forall mn \notin E$;
(ii) **Swapping:** $f_{mn}^{mk} = f_{mn}^{kn}$ for $\forall m,k,n \in V$;
(iii) **Conservation:** $I_{mn} - \Omega_{mn} = 0$ for $\forall mn \neq st$, and $I_{st} - \Omega_{st} = \lambda_{st}$, where $I_{mn} = y_{F_{mn}}(\kappa_{mn}) \cdot g_{mn} \cdot q_{mn} + q_k \sum_{k\neq m,n} f_{mn}^{mk}$ and $\Omega_{mn} = \sum_{k\neq m,n}(f_{mk}^{mn} + f_{kn}^{mn})$;
(iv) **Primitiveness:** For any $mn \in M$, either $g_{mn} > 0$, or there exists at most one $k$ such that $f_{mn}^{mk} > 0$, but not both.

□

**Explanation:** Here, $g_{mn}$ denotes the rate of entanglement generation attempts on each link $mn \in E$. To extend the pflow definition for purification, $\kappa_{mn}$ is used to denote the number of purification rounds on *mn*-ebits after successful generation. $f_{mn}^{mk}$ denotes the contribution of *mk*-ebits to generate *mn*-ebits via swapping. Condition (i) ensures the generation of elementary ebits only on physical links. Condition (ii) ensures that swapping consumes equal amounts of ebits from two source enodes. Condition (iii) maintains entanglement flow conservation by having $I_{mn} = \Omega_{mn}$ at any non-SD pair enode *mn* (*i.e.*, all generated *mn*-ebits contribute to further swapping), and defines the EDR $\lambda_{st}$ of SD pair *st*, where $I_{mn}$ and $\Omega_{mn}$ are the input and output ebit rates at enode *mn* respectively. Note that we slightly abuse the notation and define $q_{mn} = 0$ for $mn \notin E$ for notation simplicity. Finally, condition (iv) ensures that each pflow defines a *unique* way of generating end-to-end *st*-ebits by dictating that ebits between any enode *mn* can only be generated in exactly one way: either via elementary ebit generation ($g_{mn} > 0$) and/or purification (based on $\kappa_{mn}$), or via swapping at an intermediate repeater $k$ ($f_{mn}^{mk} = f_{mn}^{kn} > 0$ for at most one $k$), but not both.

*Example:* Fig. 2 shows a repeater chain with four nodes $A, B, C, D$ and SD pair *AD* without purification. While there is only one path, Fig. 2(b) shows that there are two different orders of swapping along this path. After entanglement generation along physical links, one way is to swap *AB*- and *BC*-ebits first, and then swap the successfully generated *AC*-ebits with *CD*-ebits. In the second way, *BC*- and *CD*-ebits are swapped first, and the successful ones are then swapped with *AB*-ebits. Crucially, these two ways will result in different entanglement generation and swapping rates on different links/nodes to satisfy the same EDR. Assume link generation probabilities are all 1, and swapping probabilities are uniformly 0.5. In the first way, maintaining $\lambda_{AD} = 1$ would require generation rates $g_{AB} = g_{BC} = 4$ and $g_{CD} = 2$, and swapping rates $f_{AC}^{AB} = f_{AC}^{BC} = 4$ and $f_{AD}^{AC} = f_{AD}^{CD} = 2$. But in the second way, maintaining the same $\lambda_{AD}$ would instead need generation rates $g_{AB} = 2$ and $g_{BC} = g_{CD} = 4$, and swapping rates $f_{BD}^{BC} = f_{BD}^{CD} = 2$ and $f_{AD}^{AB} = f_{AD}^{BD} = 2$. There are different ways to obtain the same EDR, leading to different generation and swapping rates at links and nodes. When the links and nodes have different costs, it is worthwhile to investigate the impact of how entanglements are established on the total cost of the entanglement users. Specifically, the pflow abstraction has interesting properties, which can be utilized to design efficient algorithms to remote entanglement distribution:

1) Each pflow corresponds to exactly one path in $G$.
2) To incorporate the purification operation, the fidelity of ebits generated along a pflow is extended based on Eq. (1):

$$F_p \triangleq \frac{1}{4}\cdot\left(1 + 3\prod W_e^{[\kappa_e]}\prod W_n\right),$$
$$\forall e \in E \cap p, n \in V \cap p, n \notin \{s,t\}. \quad (7)$$

3) Similarly, considering consumptions of purification operations, given a pflow $p$ with EDR $\lambda_{st}$, the expected cost $c_p$ for generating an end-to-end ebit is defined as follows

$$c_p \triangleq \frac{1}{\lambda_{st}}\left(\sum_{m,n,k\in p} c_k f_{mn}^{mk} + \sum_{e\in E\cap p} c_e(\kappa_e) g_e\right). \quad (8)$$

To achieve an expected EDR of $\lambda_{st}$, observe that a pflow specifies a deterministic rate of entanglement generation $g_{mn}$ on each link in each time slot; however, the actual rates of purification and swapping both depend on the rate of *successful* generation, which is a random variable. Let $c_{st}(\tau)$ be the total cost of performing all operations within one time slot. Following a given pflow $p$, the long-term expected cost is

$$c_{st} = \liminf_{T\to\infty}\frac{1}{T}\sum_{\tau=1}^{T}\mathbb{E}[c_{st}(\tau)] = c_p \cdot \lambda_{st}. \quad (9)$$

This shows that we can focus on minimizing or bounding the expected cost of maintaining a unit end-to-end EDR of 1, and then scale the cost by the actual desired EDR $\lambda_{st}$ of the SD pair. Our next focus is an algorithm for cost minimization.

**Remark: why do we need pflow?** Pflow is a crucial abstraction we adopt for our remote entanglement distribution problem, serving as the unit of a routing scheme for RED. In traditional Internet, a routing algorithm aims to find a set of network *paths*, each including a set of nodes that identifies an indivisible way for data transmission from source to destination. But pflow is more complex than a path. In quantum networks, both the nodes involved in entanglement distribution and their swapping operation order affect the number of entanglements needed on each physical quantum link relative to the expected EDR. A pflow defines both the rate of entanglements to generate on each physical link and the order of swapping operations, making it an indecomposable abstraction for entanglement distribution between source and destination.

**Remark: why pflow need to be primitive?** The primitiveness of the pflow abstraction provides two key benefits. First, since a pflow follows exactly one path with a fixed set of node pairs, the fidelity of entanglements obtained through that pflow remains constant. Second, primitiveness determines the swapping order. Swapping operations can only be performed between two node pairs after both have acquired entanglements through generation or swapping. This property ensures fixed entanglement generation and swapping rates between any node pair in the pflow, enabling us to develop a polynomial-time combinatorial algorithm for finding minimal-cost (Mi-Co) entanglement distribution in Section IV.

## IV. MI-CO: MINIMUM-COST ENTANGLEMENT DISTRIBUTION

In this section, we start with the problem of minimizing the cost for achieving a target expected EDR for a user. The problem is further complicated by non-classical swapping and purification operations, which transform routing from a node-based to a node-pair-based problem and introduce probabilistic rather than deterministic operations. However, based on a new abstraction and formulation that extends Definition 1, we will show that this problem can be solved optimally in polynomial time.

### A. Problem Description

We define the following min-cost remote entanglement distribution (Mi-Co) problem based on the pflow definition above:

*Definition 2 (Mi-Co):* Let $G = (V, E)$ be a quantum network with a source-destination pair *st*. The **minimum-cost remote entanglement distribution** problem is to find an *st*-pflow $p_{st}$ such that $c_p$ is minimized, with $\lambda_{st} = 1$.

The min-cost pflow represents the most cost-efficient way of generating a steady stream of ebits between *st* via probabilistic quantum operations. In the following, we design an optimal algorithm to solve this problem, with the help of a new abstraction called a *Purification and Swapping Graph (PSG).*

### B. Purification and Swapping Graph (PSG)

To represent all possible entanglement generation and swapping operations between quantum node pairs for RED in the quantum network, we define an auxiliary graph structure called PSG. We then design an efficient combinatorial algorithm to solve Mi-Co by finding the optimal pflow within the PSG.

*Definition 3 (PSG):* Given a quantum network $G$, its PSG is a directed multi-graph $G_{\mathsf{PSG}} = (M \cup \mathcal{V}_E, \mathcal{A})$, where each vertex in $G_{\mathsf{PSG}}$ is either an enode $m \in M$ or a virtual enode $v_{mn} \in \mathcal{V}_E$ for $mn \in E$. The edge set consists of two types of edges (arcs): $\mathcal{A} = \mathcal{A}_{\mathsf{src}} \cup \mathcal{A}_{\mathsf{swp}}$. The set $\mathcal{A}_{\mathsf{src}}$ contains arcs $a_{mn}^{[K]} = (v_{mn}, mn)$ denoting the generation and purification of ebits on each physical link $mn \in E$, with $K = 0, 1, \ldots, \mathbf{K}$; each arc has cost $c(a_{mn}^{[K]}) = c_{mn}(K)$. The set $\mathcal{A}_{\mathsf{swp}}$ contains

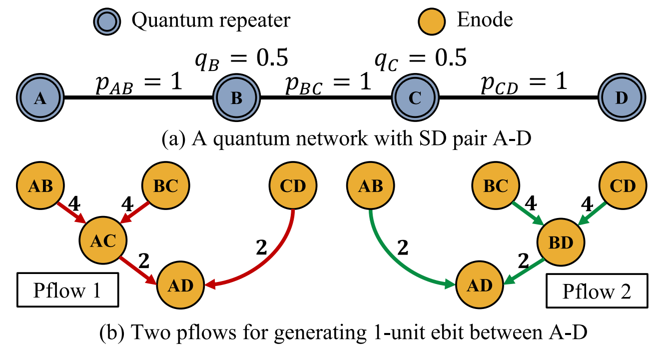



Fig. 2. Pflows (without purification) in a quantum network. Values on each pflow's arcs represent the required entanglement generation rates to get one end-to-end EDR between $AD$.

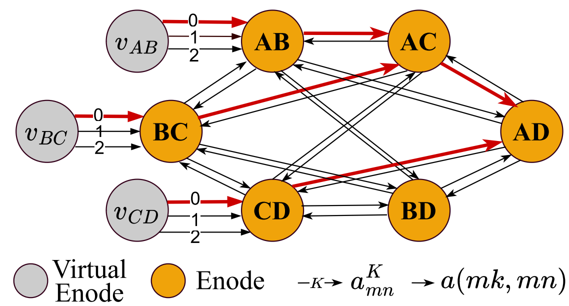



Fig. 3. An example PSG for the network in Fig. 2. Pflow 1 (with no purification) in Fig. 2 is shown as the red bold arcs.

arcs $a = (mk, mn)$ for any $m, k, n \in V$, denoting swapping contribution from *mk* to *mn* with swapping cost $c(mk, kn) = c_k/2$ so that we have $c(mk, mn) + c(kn, mn) = c_k$ which denotes the full unit swapping cost at $k$.

A PSG covers every possible enode between which ebits can be generated, and includes all possible pflows for generating these ebits as subgraphs. Specifically, a pflow is described by a *binary tree* in the PSG with special structural requirements. As in Fig. 3, a quantum network in Fig. 2 can be transformed into a PSG. The pflow in Fig. 2(b) is depicted in Fig. 3 with red bold arcs showing generation and swapping operations in the pflow. A pflow has all arcs pointing to the root enode *st*, and each enode in the binary tree has either one incoming arc from a virtual enode (generation and purification) or two matching incoming arcs from two enodes by swapping but not both.

A key advantage of the PSG is its **polynomial** size, compared to the possibly exponential number of different pflows in the network. Specifically, the PSG of a quantum network $G$ has $|V|(|V| - 1) + |E|$ enodes plus virtual enodes and $O(|V|^3 + |E|\mathbf{K})$ arcs. Note that the purification arcs $a_{mn}^{[K]}$ for $K \geq 1$ are added for generality and will only be used for fidelity consideration in Sec. V, but are not useful when solving Mi-Co since purification contributes negatively to the objective (increasing the cost). We highlight that while we utilize the PSG for assisting understanding and algorithm design, in implementation it does not need to be explicitly built. Instead, the graph structure can be implicitly incorporated into the algorithm to reduce time complexity and save memory.

**Algorithm 1** Optimal algorithm for min-cost pflow (Mi-Co)

**Input:** PSG $G_{\mathsf{PSG}} = (M \cup \mathcal{V}_E, \mathcal{A})$, SD pair $st$
**Output:** Pflow $p$ between SD pair $st$
1 $vis(v) \leftarrow 0, pa(v) \leftarrow \perp, cost(v) \leftarrow \infty, \forall v \in M \cup \mathcal{V}_E$; $cost(v) \leftarrow 0, \forall v \in \mathcal{V}_E$;
2 **while** $\exists vis(v) = 0$ **do**
3 $v \leftarrow \arg\min_{v \in M \cup \mathcal{V}_E} \{cost(v) \mid vis(v) = 0\}$;
4 $vis(v) \leftarrow 1$;
5 **if** $v = v_{mn}$ *is virtual enode in* $\mathcal{V}_E$ **then**
6 $cost(mn) \leftarrow cost(v) + \min_K \{c_{mn}(K)\}$;
7 $pa(mn) \leftarrow v_{mn}$;
8 **else**
9 **for** $k \in V$ **do**
10 **if** $vis(mk) = 0$ **then**
11 **if** $cost(mk) > (cost(mn) + cost(nk) + c(mn, mk) + c(nk, mk))/q_n$ **then**
12 $cost(mk) \leftarrow (cost(mn) + cost(nk) + c(mn, mk) + c(nk, mk))/q_n$;
13 $pa(mk) \leftarrow (mn, kn)$;
14 **if** $vis(nk) = 0$ **then**
15 **if** $cost(nk) > (cost(mk) + cost(mn) + c(mn, nk) + c(nk, mk))/q_m$ **then**
16 $cost(nk) \leftarrow (cost(mk) + cost(mn) + c(mn, nk) + c(nk, mk))/q_m$;
17 $pa(nk) \leftarrow (mn, mk)$;
18 Backtrack from enode $st$ to get the min-cost pflow $p$;
19 **return** *min-cost pflow* $p$.

### C. Minimum-Cost Pflow Algorithm Design

With the PSG abstraction, we design Algorithm 1 to solve the Mi-Co problem optimally. Line 1 initializes visited flag $vis(\cdot)$, the parent enode(s) $pa(\cdot)$ and the unit generation cost $cost(\cdot)$ for all enodes in the PSG. In each iteration, the algorithm starts from the enode with the least cost and then marks it as visited in Lines 3–4. In Lines 5–17, the algorithm checks if the newly visited enode leads to a more cost-efficient way of generating ebits for another (neighbor) enode of the visited. If a virtual enode $v_{mn}$, then the only neighbor enode *mn* of $v_{mn}$ is updated with direct generation-purification in Line 6. If *mn* is visited, then swapping is attempted for any unvisited enode *mk* or *kn*, to see if *mn-nk* or *mn-mk* can lead to less cost for *mk* or *kn*, respectively. Note how the $cost(mk)$ is updated based on the costs of source enodes $cost(mn)$ and $cost(nk)$, plus the swapping cost $c_k$, discounted by the success probability $q_n$ as in Line 12, and similarly for $cost(nk)$ in Line 16. This cost update precisely computes the cost $c_p$ as defined in Eq. (8) along with Definition 1, where $p$ is the min-cost pflow found so far for enode *mk* (and respectively *nk*). The algorithm ends when all enodes are visited. Then, based on the parent and cost sets, the *st*-pflow with minimum cost can be constructed by backtracking from the enode *st* through the parent set $pa(\cdot)$.

*Theorem 1:* Algorithm 1 computes the minimum-cost pflow for generating ebits between $s$ and $t$ in polynomial time. □

*Proof:* We prove the optimality of Algorithm 1 by induction. In the **base** case, for all enodes *mn* whose optimal way of generation is by generating elementary ebits via direct links, their costs are correctly calculated by their first updates via the virtual enodes $v_{mn}$, which all have cost 0 and hence are visited before any other enodes. In the **inductive** case, consider *mn* as the next unvisited enode with minimum cost, and let $M_{mn}$ be all enodes that have been visited so far whose minimum costs have been correctly computed by the induction assumption. Because *mn* has the minimum cost among all unvisited nodes, its current way of generation (pflow rooted at *mn*) must be optimal, otherwise, there exists at least one other enode which must have a lower cost than *mn*'s current cost to contribute to a way of generating *mn* with less cost. It follows that when any enode *mn* is visited, its current cost must be minimum. For the time complexity, since there are at most $O(|V|^2)$ enodes and $O(|V|^3)$ arcs in the PSG, the overall complexity is $O(|V|^3 \log |V|)$ using a Fibonacci heap for Line 3.

## V. CAFE: Cost-Aware High Fidelity Entanglement Distribution

Decoherence is an irreversible problem in quantum networks and has significant impacts on entanglement generation and distribution [64]. The end-to-end fidelity can decrease with imperfect quantum operations, distance, and other environmental factors. To tackle fidelity degradation, purification needs to be employed, which consumes more ebits (reducing the ebit rate) and thus increases the cost of maintaining a steady end-to-end ebit rate. Therefore, the tradeoff between cost and fidelity needs to be seriously considered.

### A. Problem Description

We formulate the cost-aware high fidelity entanglement distribution (CAFE) problem to ask: *can a stream of high-fidelity ebits be distributed at a predefined rate with bounded cost?*

*Definition 4 (CAFE):* Let $G = (V, E)$ be a quantum network with an SD pair *st*. Given a cost bound $\Delta_{st} > 0$, the **cost-aware high-fidelity entanglement distribution and purification (CAFE)** problem is to find a pflow $p$, satisfying that

1) the cost of the pflow $c_p \leq \Delta_{st}$, and
2) the fidelity $F_p$ of generated ebits is maximized.

### B. Computational Complexity and Problem Transformation

*Lemma 1:* CAFE is NP-hard. □

*Proof:* We prove NP-hardness by a reduction from the *Delay-Constrained Least-Cost (DCLC)* problem, which is NP-complete [37]. Consider an edge-weighed undirected graph $G = (V, E, \delta, \kappa)$ where each edge $e \in E$ is equipped with a nonnegative real value $\delta(e)$ to denote the *delay* of edge $e$, and a nonnegative real value $\kappa(e)$ to denote the *cost* of edge $e$. Let $(s, t)$ be an SD pair with delay bound $D$ and cost bound $C$. DCLC is to find an $s - t$ path with total cost bounded by $C$ and total delay bounded by $D$.

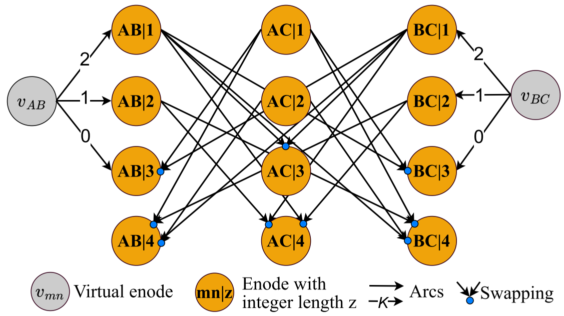



Fig. 4. LPSG for the PSG in Fig. 3 with length bound $Z_{AC} = 4$. For illustration, we set all swapping lengths $\zeta_n = 1$.

Given a DCLC instance, we construct a CAFE instance by first setting all success probabilities $q_e$, and $q_n$ to 1. We then set the link generation costs $c_e^{\mathsf{gen}} = \kappa(e)$, and node swapping costs to 0. Also set the maximum purification count to $\mathbf{K} = 0$. Then define $W_e = e^{-\delta_e}$ where $\delta_e > 0$ is the edge delay, $W_n = 1$ for each node, and a fidelity bound $\Upsilon_{st} = (1 + 3 \cdot e^{-D})/4$ and cost bound $C$. Assume we find a pflow $p$ such that $\prod_{e\in p} W_e = e^{-\sum_e d_e} \geq e^{-D}$ (satisfying the fidelity bound $\Upsilon_{st}$) and $\sum_{e\in p} c_e^{\mathsf{gen}} \leq C$, it is easy to show that the corresponding path has $\sum_e \delta_e \leq D$ and the same cost. Meanwhile, any path satisfying the delay and cost bounds also indicates a feasible pflow. Hence a solution to DCLC will give a feasible solution to this decision version of CAFE with fidelity bound $\Upsilon_{st}$ and vice versa. □

Following [32], we can transform the multiplicative node and link fidelity parameters $W_n$ and $W_e$ into additive length values $\zeta_n \triangleq -\log(W_n)$ and $\zeta_e \triangleq -\log(W_e)$ respectively. While the proof does not consider purification, we can also translate $W_e^{[K]}$ into $\zeta_e^{[K]} \triangleq -\log(W_e^{[K]})$ with $\zeta_e^{[0]} = \zeta_e$. This effectively translates the fidelity maximization problem into a length minimization problem. If a fidelity lower bound $\Upsilon_{st}$ is to be enforced, the length upper bound can be $Z_{st} \triangleq -\log\left(\frac{4\Upsilon_{st}-1}{3}\right)$.

### C. Length-Induced Layered PSG

Let us first consider a special case of CAFE, where we assume all the length values $\zeta_n$, $\zeta_e$ and $\zeta_e^{[K]}$ happen to be *positive integers*. This case is nearly (if not totally) impossible in reality, but it serves as an important building block of our algorithm design. We also assume that there is a specific fidelity (length) bound $\Upsilon_{st}$ ($Z_{st} = \lfloor -\log\left(\frac{4\Upsilon_{st}-1}{3}\right)\rfloor$). We consider the problem of finding a pflow that enforces the fidelity/length bound, while having the minimum cost. As it turns out, this problem can be solved in pseudopolynomial time (polynomial to $Z$).

Our core idea is to introduce a length-induced *Layered PSG* (LPSG), inspired by the layered graph abstraction in [51]. Given $G_{\mathsf{PSG}} = (M \cup \mathcal{V}_{\mathsf{src}}, \mathcal{A})$, we construct an LPSG $G_{\mathsf{LPSG}} = (\widetilde{M} \cup \mathcal{V}_{\mathsf{src}}, \widetilde{\mathcal{A}})$ as follows. The set $\widetilde{M}$ contains $Z + 1$ copies of each enode *mn*, denoted as $(mn|z)$ for $z = 0, 1, \ldots, Z$. Intuitively, each copy of enode $(mn|z)$ represents ebits generated between *mn* with length $z$ (corresponding to fidelity $F_{mn|z} = \frac{3e^{-z}+1}{4}$). To build the arc set, first consider generation and purification. We define arc $\tilde{a}_{mn}^{[K]} = (v_{mn}, (mn|z_{mn}^{[K]}))$, where $z_{mn}^{[K]} = \zeta_{mn}^{[K]}$ is the (integer) length value of link $mn \in E$ for $K$ rounds of purification. For swapping, we define arc $\tilde{a} = ((mk|z_1), (mn|z_2))$ for any $z_1, z_2 \leq Z$ such that $z_1 < z_2 - \zeta_k$, since swapping from *mk* with length $z_1$ into *mn* with length $z_2$ requires *kn* with length $z_3$ such that $z_1 + z_3 + \zeta_k = z_2$ and $z_3 > 0$. All so-defined arcs have the same cost as the corresponding arcs in the original PSG. Fig. 4 shows an example of LPSG generated from the subgraph in Fig. 2.

The graph is called a layered graph by viewing all vertices with the same $z$ value as one layer, and arcs crossing these layers represent fidelity changes due to generation, purification and swapping. Since the LPSG has the same structure as the original PSG, albeit with more vertices and arcs, a pflow in the original PSG will also be a pflow binary tree in this LPSG, though having its enodes spread across different layers due to the layer-crossing arcs in the LPSG. Specifically, we have the following lemma to establish the equivalence between PSG and LPSG in this special scenario with all integer values:

*Lemma 2:* For a pflow $p$ with cost $c_p$ and fidelity $F_p = \frac{1}{4}(1 + 3e^{-\zeta_p})$ where $\zeta_p \triangleq \sum_{e\in p} \zeta_e^{[\kappa_e]} + \sum_{n\in p} \zeta_n$ is the total length of the pflow satisfying $\zeta_p \leq Z$, there exists an equivalent pflow $p$ in the LPSG with the same cost, rooted at vertex $(st|\zeta_p)$, and vice versa.

*Proof:* We can use induction to show that a pflow $p \in G_{\mathsf{PSG}}$ corresponds exactly to a pflow $\tilde{p} \in G_{\mathsf{LPSG}}$ and vice versa. First, consider a pflow $p$ consisting of only a generation and/or purification arc $a_{mn}^{[K]}$. The corresponding pflow $\tilde{p}$ also contains only one arc $\tilde{a}_{mn}^{[K]}$, which points from $v_{mn}$ to $(mn|\zeta_{mn})$. By definition, Lemma 2 holds in both directions. Now consider a pflow $p$ where the root *st* is swapped from a pair *sk* and *kt* with sub-pflows $p_{sk}$ and $p_{kt}$ respectively. By induction assumption $p_{sk}$ and $p_{kt}$ have corresponding $\tilde{p}_{sk}$ and $\tilde{p}_{kt}$ rooted at layers $\zeta_{sk}$ and $\zeta_{kt}$, such that $F_{p_{sk}} = \frac{1}{4}(1 + 3e^{-\zeta_{sk}})$ and $F_{p_{kt}} = \frac{1}{4}(1 + 3e^{-\zeta_{kt}})$, respectively. Corresponding to arcs $(sk, st)$ and $(kt, st)$ in PSG are arcs $((sk|\zeta_{sk}), (st|\zeta_{st}))$ and $((kt|\zeta_{kt}), (st|\zeta_{st}))$ respectively in LPSG, with $\zeta_{st} = \zeta_{sk} + \zeta_{kt} + \zeta_k$. Such arcs must exist by the definition of the LPSG. Adding these arcs to merge $\tilde{p}_{sk}$ and $\tilde{p}_{kt}$ yields precisely a pflow $\tilde{p}_{st}$ rooted at $st|\zeta_{st}$. The opposite direction from $\tilde{p}_{st}$ to $p_{st}$ can be similarly proved. The lemma follows. □

The importance of LPSG is to translate finding a pflow with two metrics (fidelity and cost) into only one metric (cost) on a pseudopolynomial-sized graph of size $O(Z^2|V|^3)$ (in arcs). By employing Algorithm 1 with the LPSG as input, the min-cost pflow satisfying length bound $Z$ can be calculated in time polynomial to the LPSG size, as in the following theorem:

*Theorem 2:* Given a quantum network with fidelity parameters satisfying integrality conditions above and an SD pair *st* with fidelity bound $\Upsilon_{st} = \frac{1}{4}(1 + 3e^{-Z_{st}})$, the min-cost pflow can be found in time $O(Z^2|V|^3 \log(Z|V|))$.

The proof follows directly from Lemma 2 while considering a pseudopolynomial-sized graph of size $O(Z^2|V|^3)$, the time complexity of Mi-CO as proved in Theorem 1, and the size of the augmented LPSG over the original PSG.

**Algorithm 2** Approximation Scheme for CAFE

**Input:** Network $G$, SD pair $st$, cost bound $C_{st}$, accuracy $\varepsilon$
**Output:** Pflow with approximate length $Z^+$

1 Find LB and UB on $Z^*$ such that $\text{UB}/\text{LB} \le O(|V|)$;
2 **while** $\text{UB} > 4 \cdot \text{LB}$ **do** `// Stage-1`
3 $\quad Z = \sqrt{(\text{UB} \cdot \text{LB})/2}$;
4 $\quad$ Define length bound $\widetilde{Z} = 4|V| - 6$, then quantize lengths $\widetilde{\mathcal{Z}} = \mathsf{Q}_\theta(\mathcal{Z})$ where $\theta = (2|V| - 3)/Z$;
5 $\quad$ Call $A_{\mathsf{ICAFE}}$ on integer lengths $\widetilde{\mathcal{Z}}$ and bound $\widetilde{Z}$ to obtain minimum cost $\widetilde{C}$;
6 $\quad$ **if** *no feasible pflow or* $\widetilde{C} > C_{st}$ **then** $\text{LB} \leftarrow Z$;
7 $\quad$ **else** $\text{UB} \leftarrow 2Z$;
8 $\theta \leftarrow \frac{2|V|-3}{\varepsilon \text{LB}}$, $\widetilde{Z}_{\text{LB}} \leftarrow \lfloor \theta \text{LB} \rfloor$, $\widetilde{Z}_{\text{UB}} \leftarrow \lfloor \theta \text{UB} \rfloor + (2|V|-3)$;
9 Quantize all lengths $\widetilde{\mathcal{Z}} = \mathsf{Q}_\theta(\mathcal{Z})$;
10 **while** $\widetilde{Z}_{\text{UB}} > \widetilde{Z}_{\text{LB}} + 1$ **do** `// Stage-2`
11 $\quad \widetilde{Z} \leftarrow \lfloor \frac{\widetilde{Z}_{\text{LB}} + \widetilde{Z}_{\text{UB}}}{2} \rfloor$;
12 $\quad$ Call $A_{\mathsf{ICAFE}}$ on integer lengths $\widetilde{\mathcal{Z}}$ and bound $\widetilde{Z}$ to obtain minimum cost $\widetilde{C}$;
13 $\quad$ **if** *no feasible pflow or* $\widetilde{C} > C_{st}$ **then** $\widetilde{Z}_{\text{LB}} \leftarrow \widetilde{Z}$;
14 $\quad$ **else** $\widetilde{Z}_{\text{UB}} \leftarrow \widetilde{Z}$;
15 **return** *the final feasible pflow with length* $Z^+$.

### *D. An Approximation Scheme to CAFE*

Following the general principle of [32], we design an approximation scheme for CAFE. The original framework takes as input an algorithm for computing the maximum EDR subject to an (integral) fidelity bound, and provides an approximation scheme for maximizing (real-valued) fidelity subject to an EDR lower bound. We find that the same procedure can be utilized to maximize fidelity subject to a cost upper bound, and will present the corresponding algorithm which invokes the pseudo-polynomial time algorithm in the last subsection as subroutine.

Let $Z^* = -\log\left(\frac{4F^*-1}{3}\right)$ denote the length corresponding to the optimal fidelity $F^*$ achievable under a cost bound $C_{st}$, and define $A_{\mathsf{ICAFE}}$ as the minimum-cost pflow algorithm subject to an integral length bound. Also let $\mathcal{Z} = \{\zeta_n \,|\, n \in V\} \cup \{\zeta_e^{[K]} \,|\, e \in E, K = 0, 1, \ldots, \mathbf{K}\}$ denote the set of all length values. Given a quantization factor $\theta$, the function $\widetilde{\mathcal{Z}} = \mathsf{Q}_\theta(\mathcal{Z})$ quantizes each length value $\zeta \in \mathcal{Z}$ into a positive integer $\widetilde{\zeta} = \lfloor \theta\zeta \rfloor + 1$. Algorithm 2 presents our approximation scheme, which differs from [32] in 1) how to find the appropriate lower and upper bounds (as explained below), and 2) how to perform the approximate testing using $A_{\mathsf{ICAFE}}$ as a subroutine.

On Line 1 of Algorithm 2, the algorithm relies on finding a pair of initial upper and lower bounds UB and LB such that the optimal solution $Z^* \in [\text{LB}, \text{UB}]$, and that the bounds are within a polynomial factor of each other. A simple procedure can be used to find such a bound. First, sort all length values in $\zeta \in \mathcal{Z}$ in non-increasing order $\zeta_{[1]}, \zeta_{[2]}, \ldots$ in a list. Second, iterate over all values, in each iteration $i$ first pruning all arcs in the PSG with lengths larger than $\zeta_{[i]}$, and then calling Algorithm 1 to find a min-cost pflow in the remaining PSG. The procedure stops in the first iteration $j$ such that either a feasible pflow cannot be found, or the min-cost pflow has cost exceeding $C_{st}$. The value $\text{LB} = \zeta_{[j-1]}$ is thus a proper lower bound on the optimal value $Z^*$, since at least one arc with length no smaller than $\zeta_{[j-1]}$ must be kept to enforce the cost bound. Since a pflow can contain at most $|V| - 1$ links (generation/purification arcs) and $|V| - 2$ swaps, an upper bound on $Z^*$ is $\text{UB} = (2|V| - 3)\zeta_{[j-1]}$ which contains a feasible pflow satisfying the cost bound. The procedure runs in $O(|V|^3)$ time.

After finding the bounds, Algorithm 2 employs a logarithmic bisection search followed by a linear bisection to narrow down LB and UB until they are within a small bound of each other.

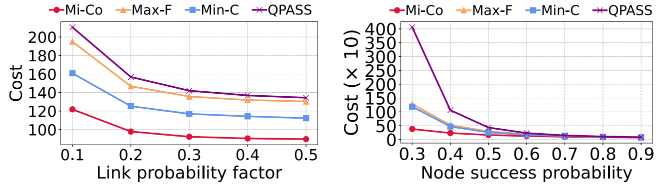
The following theorem states the performance guarantee of Algorithm 2, showing that it is a *fully polynomial time approximation scheme (FPTAS)* to the CAFE problem.

*Theorem 3:* Given $\varepsilon > 1$, Algorithm 2 finds a $(1 + \varepsilon)$-approximating pflow of the minimum length $Z^*$ subject to cost bound $C_{st}$, with time complexity $O\left(\text{poly}(|V|)\text{poly}\left(\frac{1}{\varepsilon}\right)\right)$.

*Proof:* Following the proof of Theorem 5 in [32] while being based on the new procedures for finding the initial LB and UB and Theorem 2 for finding the optimal integral solution, we first validate the near-optimality of $Z^\theta$ (minimum longest path length for quantized solution with $\theta$) in relation to $Z^*$, showing that $Z^\theta$ is tightly bounded within $\theta \cdot (1 + \varepsilon) \cdot \mathbf{Z}^*$, confirming that the initial $[\text{LB}, \text{UB}]$ framing accurately guides the approximation towards the optimal $Z^*$ without significant deviation. The quantization effectively refines the feasible solution space, ensuring that every iterative adjustment in $Z$, during the algorithmic process progressively narrows down the gap between $Z^\theta$ and $Z^*$. This iterative refinement is completed through a two-stage optimization process: When Stage-1 ends, the total number of iterations is $O(\log\log|V|)$ and each iteration calls $A_{\mathsf{ICAFE}}$ for minimum cost with at most $O(|V|^2)$ enodes and $O(|V|^3)$ arcs in the PSG. For Stage-2, the bisection is done on up to $Z_{\text{UB}} \in O\left(\frac{|V|}{\varepsilon}\right)$ integers, with up to $O\left(\log\frac{|V|}{\varepsilon}\right)$ search iterations. Each iteration calls again $A_{\mathsf{ICAFE}}$ for the minimum cost in the PSG. Since the $A_{\mathsf{ICAFE}}$ can be solved in polynomial time as proved in Theorem 2, the above time is polynomial to $|V|$ and $1/\varepsilon$. The novel use of these new subroutines as above allows our new approximation scheme to take into account both cost and purification factors, which cannot be achieved using the existing algorithms in [32]. $\square$

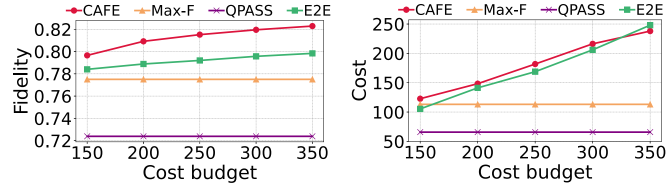
## VI. Performance Evaluation

We performed simulation-based evaluation to validate our theoretical results. We used random Waxman graphs [68] with parameters $\alpha = \beta = 0.5$. A typical inter-city quantum network has no more than 10 quantum nodes right now [4], hence we set the default graph size to 20. Graph nodes were randomly located in a synthetic 10km-by-10km area to represent the near-term city-scale quantum network setup [4]. Following existing work [32], [34], each node had an entanglement swapping success probability uniformly sampled from $[0.5, 0.75]$, and the fidelity factor $W_n$ uniformly sampled from $[0.7, 0.95]$. Our quantum link parameters were set according



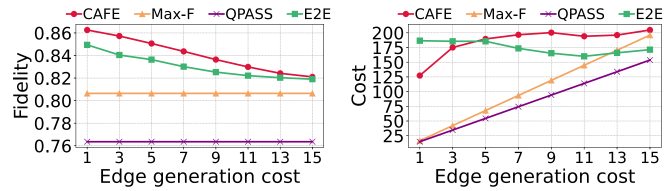

(a) Factor of the link success probability (b) Impact of node success probability

Fig. 5. Expected cost between Mi-Co and compared algorithms.




(a) Fidelity comparison (b) Cost comparison

Fig. 6. Impact of cost budget for CAFE and baselines.




(a) Fidelity comparison (b) Cost comparison

Fig. 7. Entanglement generation cost between CAFE and others.

to [20] where the link probability factor $\rho = 0.1$,[2] leading to an initial fidelity of elementary entanglements as 0.9 and $q_e$ decreasing monotonically to the link length. Due to the lack of real cost data, we aimed to validate theoretical analysis and show our algorithms' performance with fair comparison to existing works and baselines. We infer the cost based on available data to model the entanglement distribution process in a quantum network. Link generation costs were assumed to be proportional to the distance, which aligns with the existing cost for renting dark fiber, factoring in other costs like maintenance, device operations, and energy consumption [24]. Specifically, we set the unit cost for attempting entanglement generation as $5l_e$, where $l_e$ is the link length in km. The cost for performing one swapping and/or purification was 3, considering the passive quantum operations, which include one-qubit, two-qubit operations, and BSM, each costing 1 per operation. The default cost budget was set as 250. We set $\varepsilon = 0.5$ in CAFE. In each setting, we generated 10 graphs, each with 20 random SD pairs. Results were averaged over 5 runs in the same setting to average out random noise. We modify existing algorithms to consider cost as the objective.

The following algorithms were compared:

- **Mi-Co:** Algorithm 1 (without fidelity consideration).
- **CAFE:** Algorithm 2 (FPTAS to fidelity maximization).
- **Max-F:** Modified Dijkstra algorithm inspired by [41] to find the path between the source and sink with maximum fidelity without purification.
- **Min-C:** Modified Dijkstra algorithm inspired by [41] to find the path between the source and sink with minimum cost without purification.
- **QPASS:** Path-based entanglement routing in [63] modified to minimize cost.
- **E2E:** Path-based routing algorithm with link-level purification in [75], modified to satisfy the cost budget.

The following metrics were used for evaluation. The ***fidelity*** measures the fidelity values of an end-to-end entanglement generated according to the selected path or pflow. The ***cost*** measures the expected cost for generating an end-to-end entanglement with the selected path or pflow.

### A. Evaluation Results

*1) Expected Cost Comparison:* Fig. 5 shows the expected cost of Mi-Co and other algorithms with varying link success probability factors and node success probabilities. We observe that Mi-Co could always keep the lowest costs in different link success probability factors as shown in Fig. 5(a) and expected cost decreased with the increase of node success probability in Fig. 5(b). Mi-Co being an optimal algorithm showed consistent advantage in cost minimization over heuristic solutions.

*2) Impact of Cost Budget:* Figs. 6(a)–(b) show the fidelity and cost with varying cost budget. The fidelity increased with increasing cost budget in Fig. 6(a) and expected cost also increased with the increasing cost budget in Fig. 6(b). Although in Fig. 6(b), CAFE needed slightly more cost compared with others in most cases, it was well controlled within the cost budget and achieved higher fidelity in Fig. 6(a). CAFE consistently outperformed the baseline algorithms in all scenarios, by (i) employing purification which can achieve higher fidelity with higher cost budget, and (ii) finding the approximately maximum achievable fidelity under any cost budget.

*3) Impact of Entanglement Generation Cost:* Figs. 7(a)–(b) show the fidelity and cost of CAFE and other algorithms with varying generation cost. The fidelity decreased with increasing entanglement generation cost in Fig. 7(a) while performing as many purifications as possible within the cost bound in Fig. 7(b). Compared to the baselines, CAFE could always achieve the maximum fidelity while considering entanglement purification, E2E (being the only purification-aware algorithm) could sometimes incur more cost yet achieve lower fidelity.

*4) Fidelity-Cost Trade-off With Purification:* Fig. 8 shows the cost and fidelity of CAFE with varying maximum rounds for purification **K**. We observe that the fidelity and cost increased with the increase of purification rounds at the beginning but later converged when **K** became large. Two reasons led to this trend: (i) purification led to diminishing return with higher fidelity of the original ebits, and (ii) when the cost approached the cost budget, the number of actual purification rounds could not further increase while still satisfying the cost budget.

*5) Algorithm Running Time:* Fig. 9 presents the running time of CAFE and Mi-Co with different graph sizes. The line

[2]Details are in Appendix.

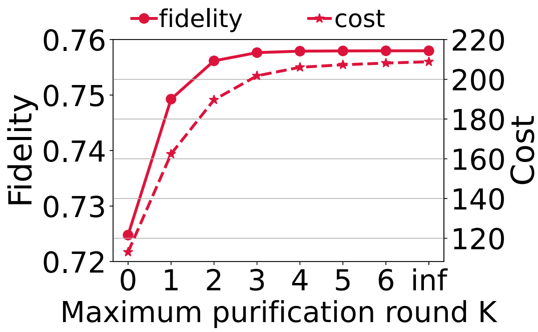
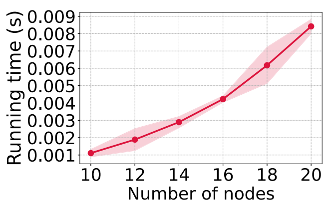
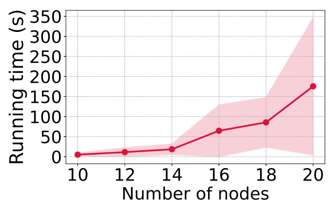



Fig. 8. Impact of purification rounds.




(a) Algorithm efficiency of Mi-Co.




(b) Algorithm efficiency of CAFE.

Fig. 9. Algorithm efficiency.

and shadow area represent the mean and standard deviation of the running time over 10 rounds with different random seeds, each testing 20 SD pairs. Routing in a quantum network is harder than in a traditional network due to it relies on entanglement swapping and teleportation between node pairs for data transmission, which leads to a large number of possible entanglement scenarios. A quantum network with 20 nodes contains 190 different node pairs, and probabilistic purification on node pair with quantum links further complicates the problem. From Fig. 9 we notice that the algorithm running time increases as the quantum network scales up. However, in a quantum network with 20 nodes, Mi-Co completes in under 10ms, while CAFE solves on average within 3 minutes. Importantly, these algorithms only need to execute once, with all subsequent operations following the established rules. This minute-level algorithm running overhead is acceptable for quantum networks with stable topologies. Even if a sudden failure that causes topology changes occurs, the algorithm can restore the near-optimal RED within a short time.

To summarize, we make three key observations from the evaluation: (i) Using the pflow abstraction, **Mi-Co can achieve the optimal cost** for distributing entanglement, lower than the lowest cost achievable by any path-based algorithm; (ii) When fidelity is considered, **link-level purification is important to achieve high fidelity** that can almost always outperform algorithms without purification; (iii) **CAFE can consistently outperform path-based heuristic algorithms** due to theoretical guarantee based on the pflow and PSG/LPSG abstractions.

## VII. Related Work

Early work in quantum networking focused on feasibility demonstration in ideal situations. Schoute et al. [61] presented efficient routing algorithms in ring and sphere topologies. Pant et al. [55] developed and analyzed entanglement routing in a square-grid topology. The limitation of these is the focus on special network topologies that are hardly realistic in practice.

Recent works have focused on general quantum networks. Shi and Qian [63] proposed Q-PASS and Q-CAST for entanglement routing to maximize throughput. Zhao and Qiao [74] designed a redundant entanglement provisioning and selection algorithm to maximize throughput. Dai et al. [18] proposed the first optimal remote entanglement distribution (ORED) protocol to achieve the maximum end-to-end throughput with long-term quantum memories. In addition to the rate, the fidelity of distributed entanglements has recently been studied. Zhao et al. [75] designed a fidelity-aware routing and purification (EFiRAP) protocol in bit flip errors to maximize throughput.

As pointed out, the above works ignore the critical cost factor in quantum network protocol design, which was the main driving force behind the Internet. Some works have considered the energy efficiency of quantum communication. For instance, Galve and Lutz [30] investigated the energy cost of entanglement production, which scales exponentially with the amount of entanglement in the harmonic chain. Beny et al. [8] also quantified the energy cost of entanglement extraction in quantum field vacua which grows exponentially with extracted entanglement if the cost depends on the spatial dimension. Meier and Yamasaki [50] proposed a general framework for studying the energy consumption of quantum and classical computation. Jaschke and Montangero [40] theoretically showed the potential quantum advantage in energy efficiency from the quality of quantum gates and the entanglement generated in the quantum processing unit. However, these analyses are neither comprehensive nor directly related to quantum networking design. At the same time, they ignore the service price from end users instead of service providers.

Our work seeks to fill this gap by introducing the first set of optimal and approximation algorithms to explore the trade-off between cost, entanglement rate and fidelity in entanglement distribution. Our approach is inspired by ORED above [18], [19], as well as FENDI [32], a recently proposed novel approximation scheme for achieving high-fidelity entanglement distribution with entanglement rate guarantee. Specifically, FENDI is built upon the *eflow* abstraction in ORED [18], and proposes a new *pflow* abstraction and a novel decomposition theorem to achieve approximately maximum worst-case fidelity while satisfying entanglement rate requirement with entanglement generation and swapping. However, both ORED and FENDI do not consider purification, which is of central importance in improving fidelity in an actual quantum network. They also do not have any cost-related consideration. This paper builds upon the abstractions developed in ORED and FENDI, and extends them to a new graph-based abstraction that (1) incorporates purification as part of the entanglement distribution process, and (2) considering cost as a primary optimization constraint due to its practical importance, in addition to entanglement rate. These two considerations make the problem that we study significantly more challenging than ORED or FENDI, and to our knowledge there is no existing work that can provide cost-efficient distribution, high-fidelity entanglement distribution with strong theoretical guarantee as our work.

## VIII. Discussion

**Quantum resources:** Quantum resources such as quantum memory capacity are scarce in existing and near-future quantum networks, while current research aims to develop new hardware [45], [62] to resolve. Recognizing this limitation, we assign costs for quantum resources used in entanglement distribution in this work. Rather than imposing hard constraints, we convert the generation of entangled photon pairs and the construction and maintenance of quantum channels into cost values. Quantum memory consumption and probabilistic losses during quantum operations are also expressed in terms of costs, where more entangled photons require more memory slots, and higher loss probability requires more entanglement generation. Envisioning near-future quantum networks, providers can assign costs to resources according to their scarcity, transforming availability constraints into cost considerations while reducing demand for expensive resources. This economic model mirrors approaches widely used in Internet economics [38], [47], [49], [69]. For example, reference [47] sets a shadow price proportional to the inverse of available resource capacity, derived from the first-order condition of a network utility maximization program. The more resources are available, the lower the prices are for a user. In [47], the resource is typically the classical data rate. In our quantum network, the resource can be defined as the maximum entanglement throughput achievable over each link, given available physical resources like quantum memory, optical bandwidth, etc. When extending to multiple SD pairs, a price bidding campaign can be launched before incorporating new SD pairs [38], enabling fair and controllable quantum network operation with limited quantum resources.

**Tackling multiple SD pairs:** This paper focuses on optimizing the cost for fulfilling a single user request (an SD pair), from the view of a cost-minimizing network operator or the user. If the network has enough resources to handle many SD pairs without bottleneck, then our algorithms can be easily extended to the multi-SD pair case by solving for each SD pair independently. When resources are not abundant, resource contention occurs among SD pairs. There are two ways to address this conflict. First, quantum network service providers can implement time-slice sharing of the network among SD pairs. For each SD pair, they can allocate resources to each pair using the methods proposed in this work. In this approach, each SD pair receives a specific deadline for network usage in a certain period. Following the algorithm in [34], we can schedule each SD pair with additional cost budget constraints to the multi-SD-pair entanglement distribution scheduling problem. Second, service providers can set prices based on resource consumption and bidding results from all SD pairs, using cost as an intermediate metric for resource allocation [66]. Our algorithm may specifically contribute to multi-SD pair entanglement distribution in one promising direction. By launching a bidding campaign and dynamically setting costs for links and nodes based on current resource contention and users' willingness to pay, network operators can employ our algorithms to develop competitive online solutions that maximize the number of accepted user requests, following established online algorithm design frameworks [10], [38].

**Quantum network abstraction:** Early works [63], [74], [75] on entanglement distribution in quantum networks focused on finding optimal end-to-end paths. Unlike classical networks, quantum networks require swapping between entangled qubit pairs to establish end-to-end entanglements. To better represent this unique architecture, researchers proposed abstractions such as pflow [32] and swapping tree [31]. In this work, we incorporate purification into pflow in Section III-D and propose a network-level abstraction called PSG that goes beyond both pflow and the swapping tree. PSG encompasses all possible entanglement generation, physical-link-level purification, and swapping for entanglement distribution between specific pairs of quantum nodes. Compared to existing abstractions [31], PSG incorporates cost considerations and includes purification for high-fidelity entanglement distribution. Furthermore, we demonstrate that PSG maintains polynomial size relative to the number of nodes in the original quantum network, making it a practical abstraction for solving cost-optimal entanglement distribution problems.

**Order of purification and swapping:** The sequence of purification and swapping for high-fidelity entanglement distribution has been explored in existing work [54]. Following their research, which shows that purification-before-swapping achieves a higher rate and fidelity, we adopt the same operational sequence. Our proposed algorithms can also accommodate swapping-before-purification by replacing the root enode of the SD pair with a virtual enode and adding arcs for purification from the virtual enode to an actual root enode. We leave the exploration of such an operational sequence for future work.

## IX. Conclusion

In this paper, we designed cost-efficient entanglement distribution protocols in a general optical quantum network. We started with the problem of establishing a stream of entanglements with minimum cost and designed an optimal algorithm based on a new abstraction called the Purification and Swapping Graph (PSG). We then formulated the cost-aware high fidelity entanglement generation (CAFE) problem and proved its NP-hardness. Based on a novel extension to the PSG, we developed an approximation scheme, which can find an entanglement distribution protocol that achieves an end-to-end fidelity arbitrarily close to the optimal achievable fidelity for maintaining a stream of entanglement distribution with bounded cost. Extensive simulation results showed the superior performance of our algorithms in terms of cost efficiency and fidelity compared to existing algorithms and heuristic solutions.

## Appendix
## Explanation of the Entanglement Success Probability

Following the existing work [20], we have

$$q_e = 1 - \left[1 - p_{succ}\left(1 - p_{tloss}\right)\right]^{N_{attp}}, \tag{10}$$

where $p_{succ}$ is the entanglement source's generation efficiency, $p_{tloss}$ is the probability of transmission loss, and $N_{attp}$ is the number of attempts in unit slot. For a source, its efficiency and the fidelity[3] are balanced by a link probability factor $\rho \in [0, 1]$

$$p_{succ} = \rho \times 10^{-3}, \tag{11}$$

and $p_{tloss}$ increases exponentially with the link length $l_e$ as

$$p_{tloss} = 1 - 10^{-\frac{l_e \times 0.1\text{dB/km}}{10}}. \tag{12}$$

Considering the delays of qubit operation, photon emission, and electron readout, $N_{attp}$ can be represented as

$$N_{attp} = \frac{t_{mem} - t_{op}}{2l_e/c + t_{em} + t_{read}}, \tag{13}$$

where $c \approx 3 \times 10^8$m/s is the light speed, $t_{mem} = 1.46$ seconds is the maximum entangled time for NV-based quantum memory, and $t_{op} = 1.1 \times 10^{-9}$, $t_{em} = 5.5 \times 10^{-6}$, $t_{read} = 3.7 \times 10^{-6}$ are qubit operation, photon emission, and electron readout delays in seconds, respectively [17] and [20].

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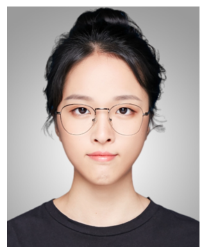
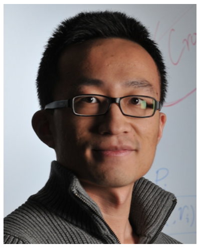


**Huayue Gu** (Graduate Student Member, IEEE) received the B.E. degree in computer science from Nanjing University of Posts and Telecommunications, Jiangsu, China, in 2019, and the M.S. degree in computer science from the University of California at Riverside, CA, USA, in 2021. She is currently pursuing the Ph.D. degree with the Computer Science Department, North Carolina State University. Her research interests include quantum networking, quantum communication, and data analytics.

**Zhouyu Li** (Graduate Student Member, IEEE) received the B.E. degree from Central South University, Changsha, China, in 2019, and the M.S. degree from Georgia Institute of Technology, Atlanta, USA, in 2020. He is currently pursuing the Ph.D. degree in computer science with North Carolina State University. His research interests include privacy, cloud/edge computing, and network routing.

**Xiaojian Wang** (Graduate Student Member, IEEE) received the Ph.D. degree in computer science from North Carolina State University, USA, in 2025. She is currently an Assistant Professor of computer science and engineering with the University of Colorado Denver, USA. Her research interests include satellite networks, edge computing, security, blockchain, and payment channel networks. She is a member of the Technical Committee of INFOCOM WKSHPS PerAI-6G 2022–2025, ACM CCS 2024 Artifact Evaluation, and INFOCOM 2026. She has served or is serving on the organizing committees of IEEE IPCCC 2023–2025.

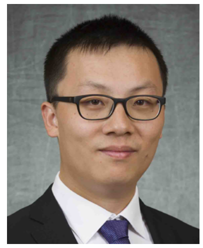


**Dejun Yang** (Senior Member, IEEE) received the B.S. degree in computer science from Peking University, Beijing, China, in 2007, and the Ph.D. degree in computer science from Arizona State University, Tempe, AZ, USA, in 2013.

Currently, he is an Associate Professor of computer science with Colorado School of Mines, Golden, CO, USA. His research interests include the Internet of Things, networking, and mobile sensing and computing, with a focus on the application of game theory, optimization, algorithm design, and machine learning to resource allocation, security, and privacy problems.

Prof. Yang has received the ACM SACMAT Test of Time Award in 2024, the IEEE Communications Society William R. Bennett Prize in 2019 (Best Paper Award for IEEE/ACM TRANSACTIONS ON NETWORKING and IEEE TRANSACTIONS ON NETWORK AND SERVICE MANAGEMENT in the previous three years), and the Best Paper Awards at the IEEE Global Communications Conference (GLOBECOM) in 2015, the IEEE International Conference on Mobile Ad hoc and Sensor Systems (MASS) in 2011, and the IEEE International Conference on Communications (ICC) in 2011 and 2012. He received the Best Paper Award Runner-Up at the IEEE International Conference on Network Protocols (ICNP) in 2010. He was the Symposium Co-Chair of the International Conference on Computing, Networking and Communications (ICNC) in 2016 and 2025. He is the TPC Vice Chair of Information Systems for the IEEE International Conference on Computer Communications (INFOCOM) in 2020 and the Student Travel Grant Co-Chair of INFOCOM 2018–2019. He serves as an Associate Editor for IEEE TRANSACTIONS ON MOBILE COMPUTING, IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, and IEEE INTERNET OF THINGS JOURNAL.

**Guoliang Xue** (Fellow, IEEE) is currently a Professor of computer science with the School of Computing and Augmented Intelligence, Arizona State University. His research interests include the Internet of Things, cloud/edge/quantum computing and networking, crowdsourcing and truth discovery, QoS provisioning and network optimization, security and privacy, optimization, and machine learning. He received the IEEE Communications Society William R. Bennett Prize in 2019. He has served as the VP-Conferences of the IEEE Communications Society. He is the Steering Committee Chair of IEEE INFOCOM. He is an Associate Editor of IEEE TRANSACTIONS ON MOBILE COMPUTING and a member of the Steering Committee of this journal. He served on the editorial boards for IEEE/ACM TRANSACTIONS ON NETWORKING and *IEEE Network Magazine* and the Area Editor for IEEE TRANSACTIONS ON WIRELESS COMMUNICATIONS, overseeing 13 editors in the wireless networking area.



**Ruozhou Yu** (Senior Member, IEEE) received the Ph.D. degree in computer science from Arizona State University, USA, in 2019. He is currently an Assistant Professor of computer science with NC State University, USA. His interests include quantum networking, edge computing, algorithms and optimization, distributed learning, and security and privacy. He received the NSF CAREER Award in 2021. He served on the organizing committees of IEEE INFOCOM 2022–2025, IEEE IPCCC 2020–2024, and IEEE QCNC 2024, including as the General Chair for IPCCC 2024, the TPC Chair for IPCCC 2023, the TPC Track Chair for IEEE ICCCN 2023, and a TPC Member for IEEE INFOCOM 2020–2026, ACM Mobihoc 2023–2025, and IEEE ICDCS 2025, among others. He serves as an Area Editor for *Computer Networks* (Elsevier) and an Associate Editor for IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, who won the TNSE Excellent Editor Award in 2024.